

MTH 446: Lecture # 12

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Lecture Span

1. Complex derivatives

Complex derivatives

Note, by the necessary condition of complex differentiability, f' DNE for z such that the CR equations are not satisfied.

By the sufficient condition of CD, f' exists for z such that the CR equations are satisfied.

CR equations in polar coordinates

$$u_r = \frac{1}{r} v_\theta \quad (1)$$

$$\frac{1}{r} u_\theta = -v_r \quad (2)$$

Proof. Use the chain rule and the fact that $x = r \cos \theta$ and $y = r \sin \theta$ to convert the CR equations in rectangular coordinates to polar coordinates. That is,

$$\frac{du}{d\theta} = u_x x_\theta + u_y y_\theta \quad (3)$$

□

$$u_x = u_r \cos \theta + \sin \theta / r \cdot u_\theta \quad (4)$$

$$u_y = u_r \sin \theta - \cos \theta / r \cdot u_\theta \quad (5)$$

Example

If $z_0 \neq 0$, then $f'(z_0)$ exists, then

$$f'(z_0) = -\frac{i}{z_0} (u_\theta + i v_\theta) \quad (6)$$

Proof. $f'(z_0) = u_x + i v_x = u_r \cos \theta - \sin \theta / r \cdot u_\theta + i(v_r \cos \theta + \sin \theta / r \cdot v_\theta)$. You then obtain

$$= -\frac{i}{r} [u_\theta (\cos \theta - i \sin \theta) + i v_\theta (\cos \theta - i \sin \theta)] \quad (7)$$

$$= -\frac{i}{r} [(u_\theta + i v_\theta) e^{-i\theta}] = -\frac{i}{r e^{i\theta}} (u_\theta + i v_\theta) = -\frac{i}{z_0} (u_\theta + i v_\theta) \quad (8)$$

□

More Operators

Will not be used in class, but useful to mention for completeness.

$$\partial_z = \frac{1}{2} (\partial_x - i\partial_y) \quad (9)$$

$$\partial_{\bar{z}} = \frac{1}{2} (\partial_x + i\partial_y) \quad (10)$$

$$(11)$$

Note that

$$\partial_{\bar{z}} f = 0 \iff \text{CR equations} \quad (12)$$

$$\partial_z f = u_x + iv_x \quad (13)$$

Analytical Functions

Definition 0.1. Let D be open in $\mathbb{C}$ and not empty. Then, f is analytic at z_0 if there is $\epsilon > 0$ (NOT FOR ALL $\epsilon > 0$) such that $D_\epsilon(z_0) \subseteq D$ and f' exist for every $z \in D_\epsilon(z_0)$.

Note, if f' is only differentiable at z_0 but DNE for some z close to z_0 , then f is not analytic at z_0 .

Definition 0.2. f is analytical on U if it is analytic at every point U , which is the domain of f . Set of all analytical functions is denoted by $H(U)$, H meaning holomorphic.

Theorem 0.3. Let f and g analytic, then

1. $\alpha f + \beta g \in H(U)$ is analytic
2. fg is analytic.
3. f/g is analytic if $g \neq 0$ at z_0 .

Theorem 0.4. Uniqueness property of analytic functions.